VIETNAM NATIONAL UNIVERSITY HO CHI MINH CITY
HO CHI MINH CITY UNIVERSITY OF TECHNOLOGY

Vietnamese Food Classification with CNN and ResNet-50 on 30VNFoods

CO3133 Deep Learning and Its Applications

Group NNs | Semester 261 | Instructor: Lê Thành Sách

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HO CHI MINH CITY · OCTOBER 2026

CO3133 / ASSIGNMENT 2

Vietnamese Food Classification

CNN and ResNet-50 on 30VNFoods

DEEP LEARNING AND ITS APPLICATIONS

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e 3: A training batch after preprocessing. The input tensor has shape $16 \times 3 \times 224 \times 22$

CO3133 / ASSIGNMENT 2

Presentation roadmap

01 **Problem & Data**

Motivation · Dataset · Cleaning

02 **Methodology**

CNN · ResNet-50 · Training protocol

03 **Experimental Results**

Accuracy · Macro-F1 · Ablation

04 **Analysis & Conclusion**

Failures · Grad-CAM · Limitations

01 / PROBLEM AND DATA

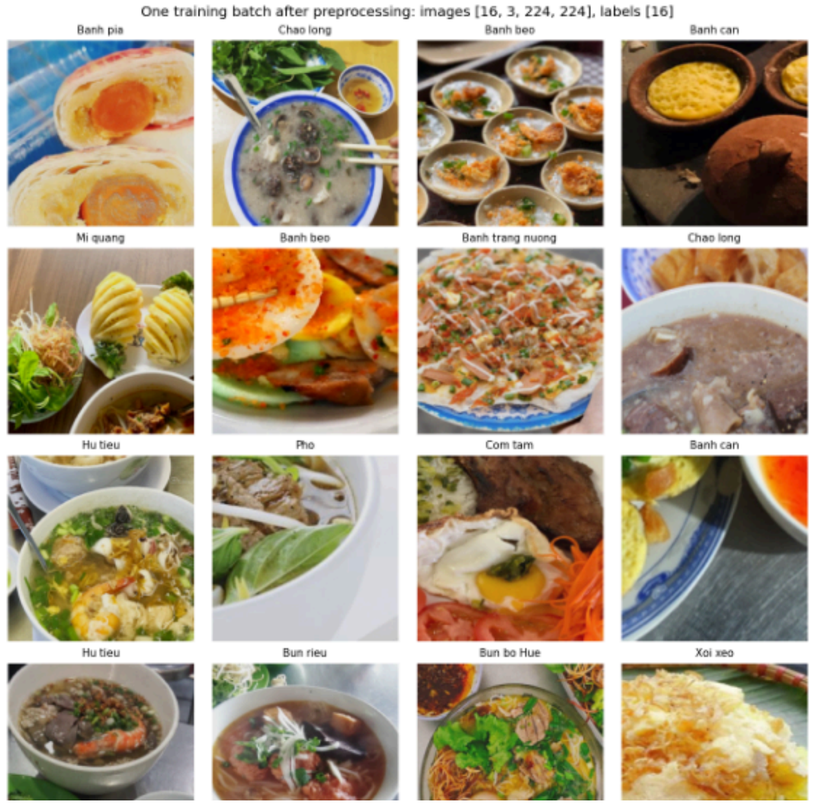
Problem motivation & research question

Why Vietnamese food classification?

Dishes may share broth, toppings, wrappers and presentation styles.

Primary research question

Does adapting ImageNet-pretrained features outperform training only a classification head?



Controlled comparison: frozen ResNet-50 vs full fine-tuning; scratch CNN as reference.

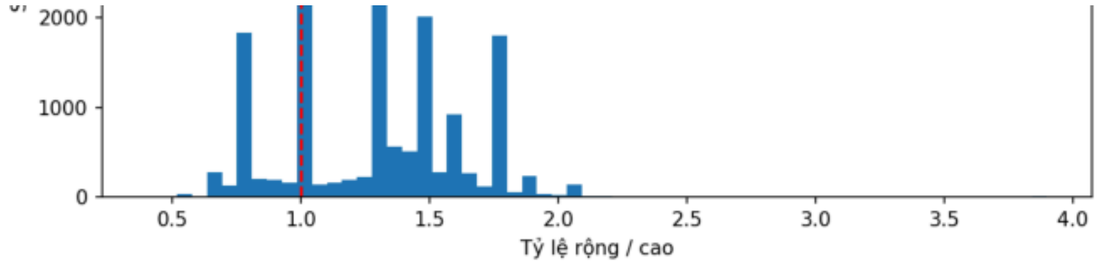
01 / PROBLEM AND DATA

30VNFoods: dataset overview

30
Food categories

25,136
Original images

224 × 224
Model input size



igure 2: Width-to-height ratios in the original training partition. The dashed line marks square nputs.

ı a fixed-seed sample of 3,000 original training images, class distributions of mean RGB intensity verlapped. Banh pia had the highest median intensity in that sample. Visual inspection ound examples showing pastries with packaging on white backgrounds. Photos also show that

Dataset challenges

Class imbalance, variable image dimensions, ambiguous presentation and potential duplicate leakage.

01 / PROBLEM AND DATA

Data cleaning & leakage prevention

Original dataset
25,136 images

→

MD5 + 64-bit pHash
Exact / potential near-duplicates

→

Duplicate graph
Connected components

→

Filter conflicts / copies
507 images removed

→

Cleaned split
24,629 images

Original → Cleaned partitions

TRAIN	17,581	→	**17,164**
VALIDATION	2,515	→	**2,481**
TEST	5,040	→	**4,984**

LIMITATION

Visually similar but distinct images may be incorrectly grouped by pHash and removed.

Goal: reduce cross-split leakage and label conflicts, not simply maximize dataset size.

01 / PROBLEM AND DATA

Image preprocessing & augmentation

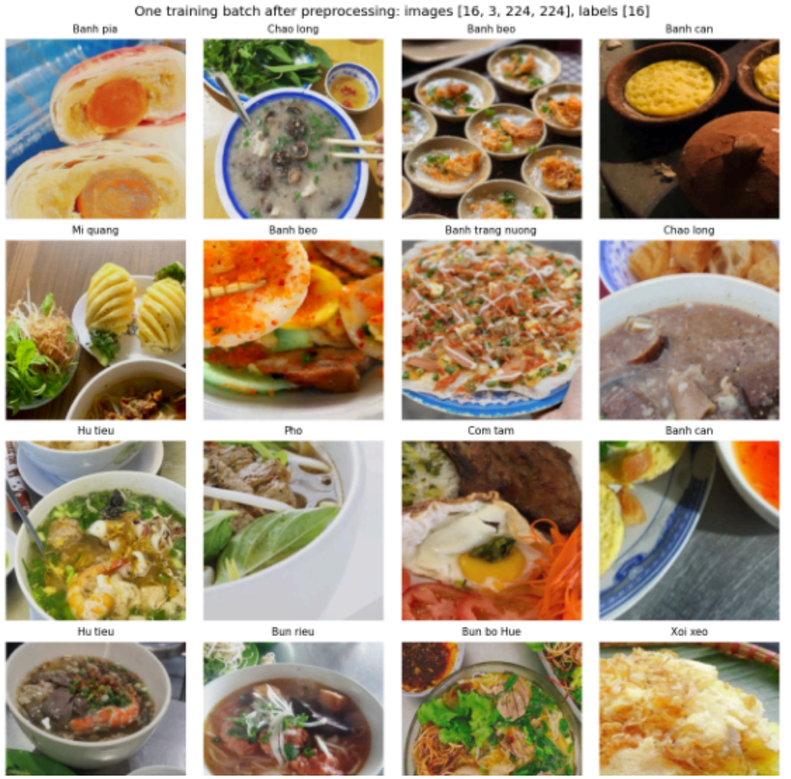
e 3 shows augmented crops with their dataset labels. Normalization is reversed for displa

Training transforms

RandomResizedCrop (224×224), horizontal flip, color jitter (0.1).

Validation / test transforms

Resize shortest side to 256; center crop 224×224; no random augmentation.



CNN: training-set normalization. ResNet: ImageNet normalization. Batch input [B, 3, 224, 224].

02 / METHODOLOGY

End-to-end experimental pipeline

Cleaned split	Train 3 models	Validate	Save best.pt	Evaluate test
17,164 / 2,481 / 4,984	CE loss + Adam	Macro-F1, patience 5	Best validation checkpoint	Metrics + error analysis

Training

Backpropagate and update parameters

Test

Restore best.pt; no checkpoint selection on test

02 / METHODOLOGY · ARCHITECTURE

SmallCNN baseline — architecture

RGB input
3 × 224 × 224

›

Block 1
32 × 112 × 112

›

Block 2
64 × 56 × 56

›

Block 3
128 × 28 × 28

›

Block 4
256 × 14 × 14

›

GAP + FC
256 → 30

Each convolutional block
Conv 3×3 → BatchNorm → ReLU → MaxPool 2×2

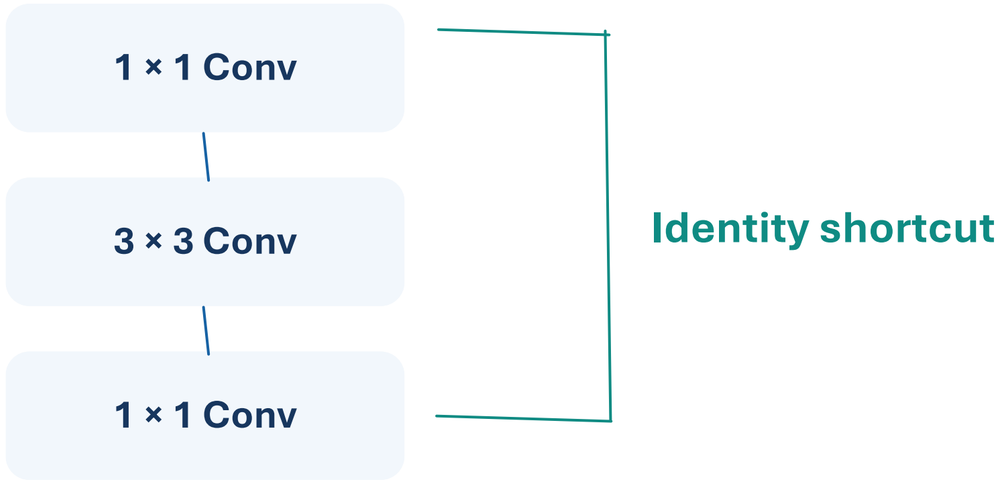
Classifier
Global Average Pooling → Dropout 0.3 → Linear 256→30

397,086 trainable parameters · trained from scratch · single seed (42).

02 / METHODOLOGY · ARCHITECTURE

Why ResNet-50? Residual learning

Residual bottleneck



Key idea

Add the input (shortcut) to learned residual features. This supports effective training of deeper networks.

ImageNet-1K V2 pretrained weights

02 / METHODOLOGY · ARCHITECTURE

ResNet-50 — model architecture



Backbone

Residual bottleneck stages [3, 4, 6, 3]. Final feature maps: 2048 × 7 × 7.

New classification head

Linear 2048 → 30 classes, with 61,470 parameters.

Total parameters: 23,569,502 in both ResNet variants.

02 / METHODOLOGY · COMPARISON

Frozen backbone vs full fine-tuning

FROZEN BACKBONE

ResNet feature extractor

No gradient updates · BatchNorm eval

Train only classifier head

Head LR = 1e−3

61,470 trainable

FULL FINE-TUNING

ResNet feature extractor

Backbone updates · BatchNorm train

Train backbone + classifier

Backbone LR = 1e−4; Head = 1e−3

23,569,502 trainable

02 / METHODOLOGY

Training setup & evaluation protocol

SETTING	CNN	RESNET FROZEN	RESNET FULL
Training seed(s)	42	42, 43, 44	42, 43, 44
Batch size	64	32	32
Maximum epochs	20	15	15
Early stop patience	5	5	5
Head learning rate	1e−3	1e−3	1e−3
Backbone learning rate	—	frozen	1e−4

Cross-entropy loss + Adam (weight decay 1e−4); checkpoint selected by validation macro-F1.

02 / METHODOLOGY

Metrics and checkpoint selection

Primary metric · Macro-F1

Average classwise F1 across all 30 categories.

Secondary metric · Accuracy

Fraction of correctly predicted test images.

Top-3 Accuracy

Whether the true dish appears among three highest-scoring classes.

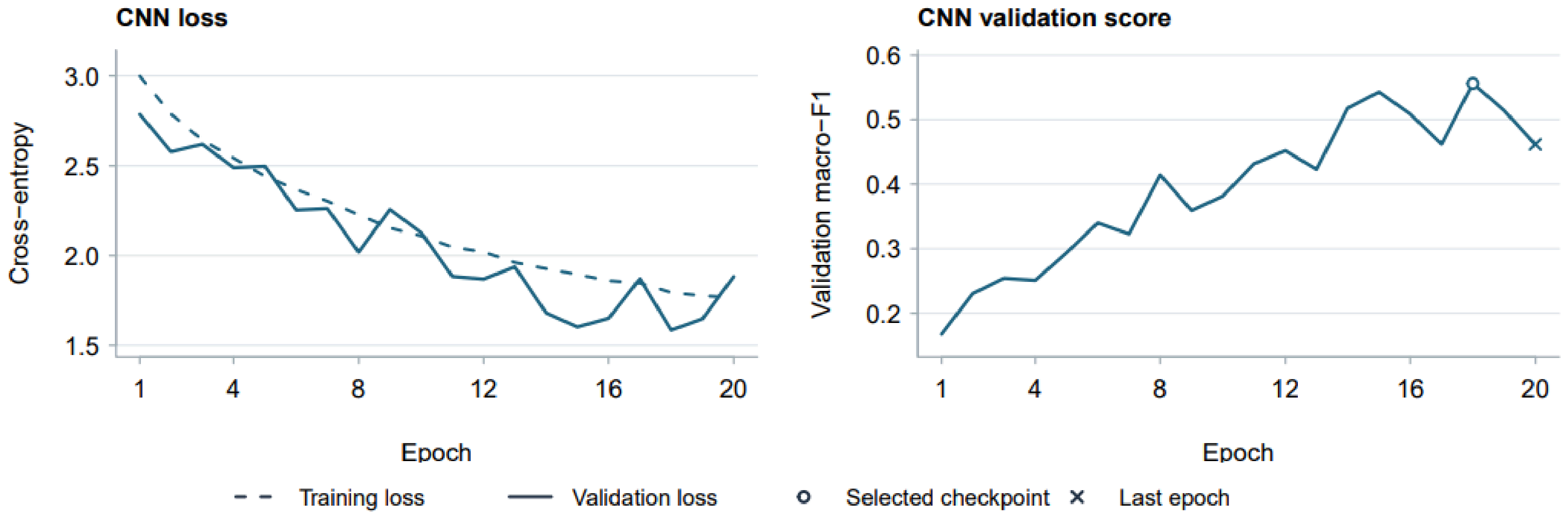
Checkpoint rule

Save best.pt only when validation macro-F1 improves; patience = 5.

Held-out test set: 4,984 images, used after validation-based checkpoint selection.

03 / EXPERIMENTAL RESULTS

Training dynamics



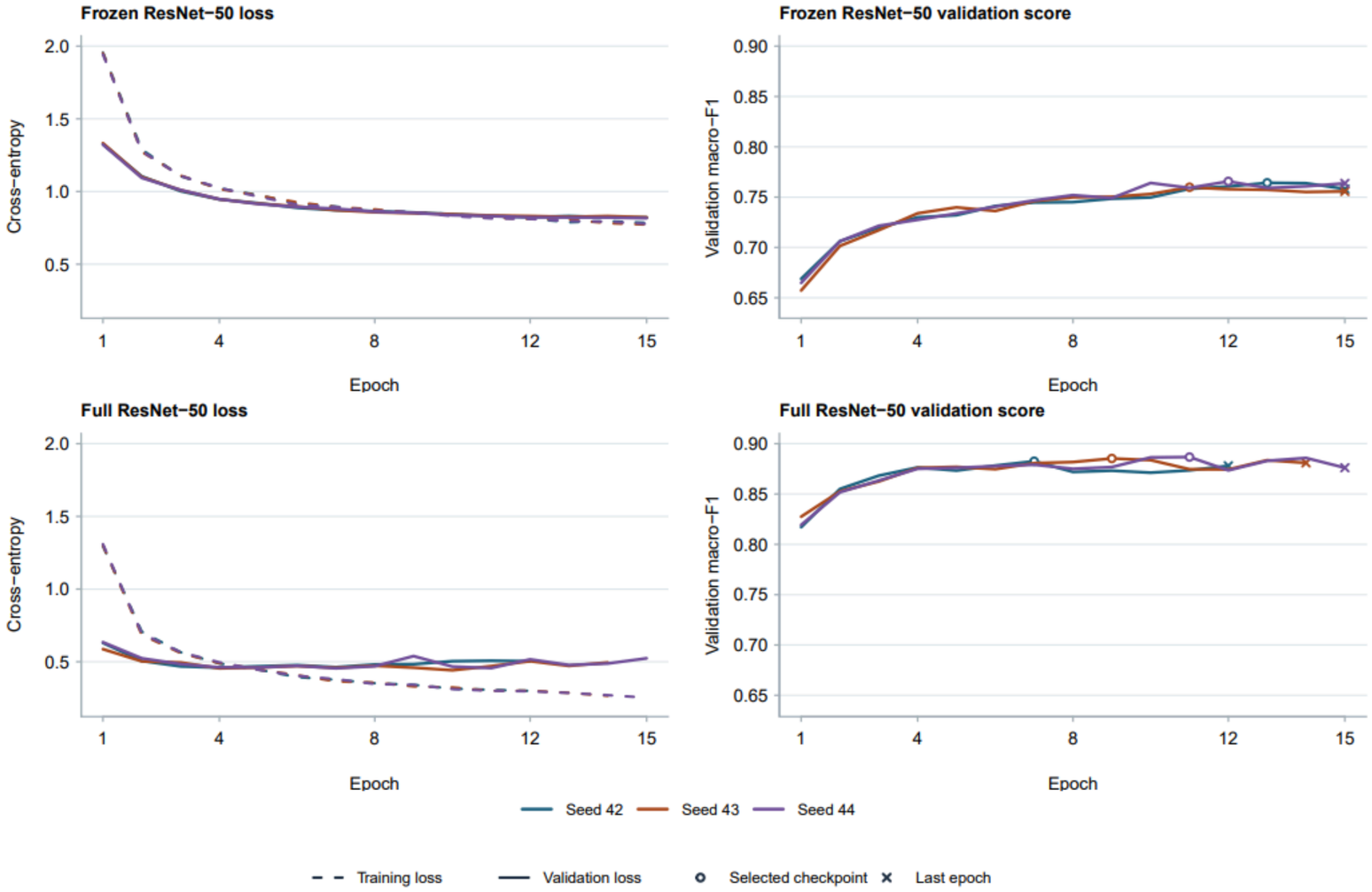
CNN baseline

Frozen plateaus near 0.76 validation macro-F1; full fine-tuning peaks around 0.88–0.89.

03 / EXPERIMENTAL RESULTS

Training dynamics

Three paired seeds — ResNet-50



Frozen plateaus near 0.76 validation macro-F1; full fine-tuning peaks around 0.88–0.89.

03 / EXPERIMENTAL RESULTS

Main test results

Test accuracy



Full fine-tuning: +12.29 percentage points accuracy and +0.1202 macro-F1 vs frozen.

Macro-F1: CNN 0.5403 · Frozen 0.7720 ± 0.0022 · Full 0.8922 ± 0.0063.

03 / EXPERIMENTAL RESULTS

Controlled ablation across three seeds

SEED	FROZEN MACRO-F1	FULL MACRO-F1	GAIN
42	0.7698	0.8849	**+0.1150**
43	0.7720	0.8957	**+0.1236**
44	0.7742	0.8961	**+0.1219**
MEAN	**0.7720**	**0.8922**	**+0.1202**

All three matched seeds favor full fine-tuning; the smallest paired gain is +0.1150 macro-F1.

03 / EXPERIMENTAL RESULTS

Accuracy vs computational cost

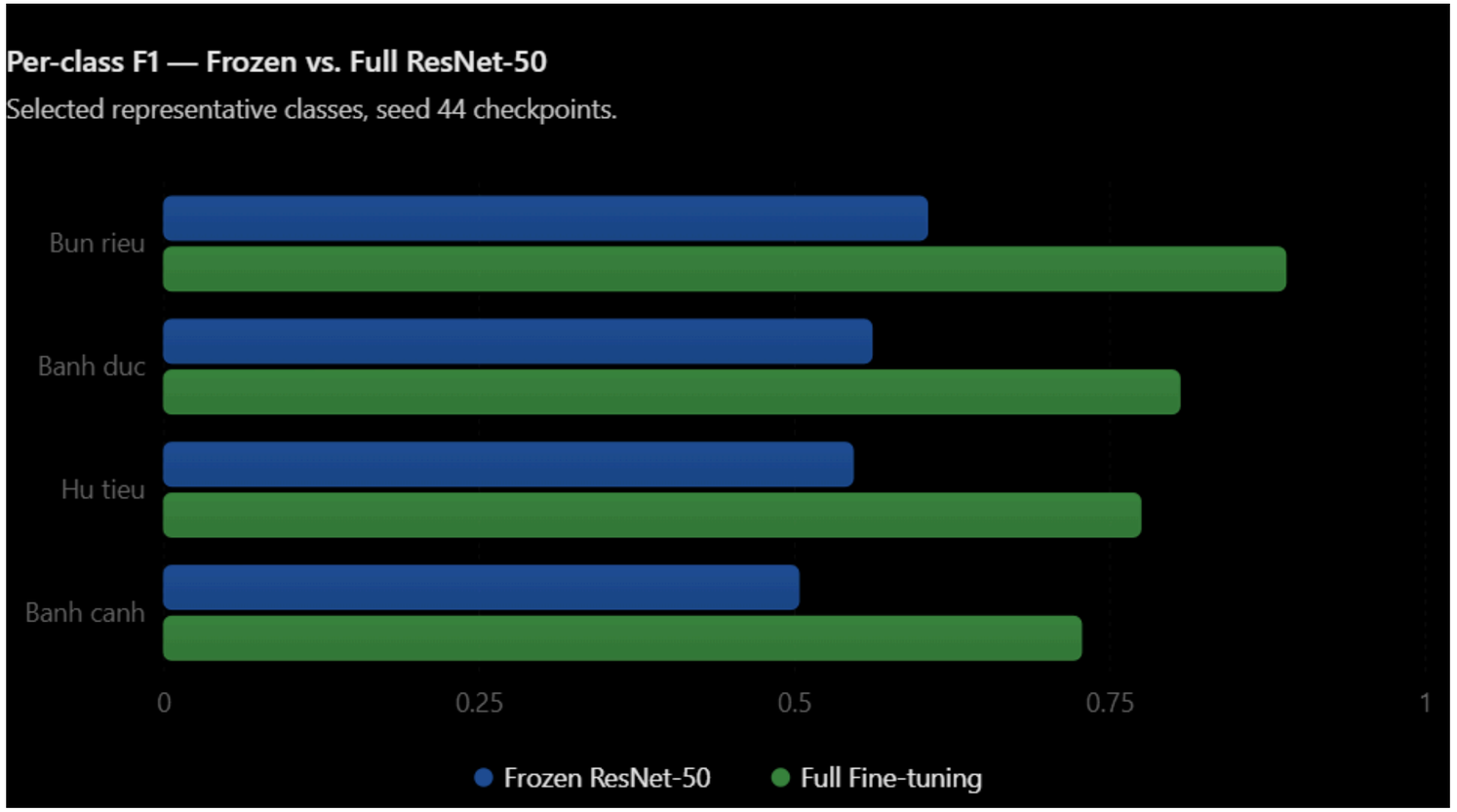
MODEL	TRAINING TIME	PEAK TRAIN VRAM	FORWARD / IMAGE
CNN scratch	20.85 min	Not recorded	0.859 ms
ResNet frozen	12.13 min	0.434 GiB	4.312 ms
ResNet full	24.97 min	2.894 GiB	4.285 ms

Full: ~2.06× training time and ~6.67× peak GPU memory vs frozen.

Training: RTX 4060 Laptop GPU · Forward timing: Apple MPS (batched, excludes data loading).

03 / EXPERIMENTAL RESULTS

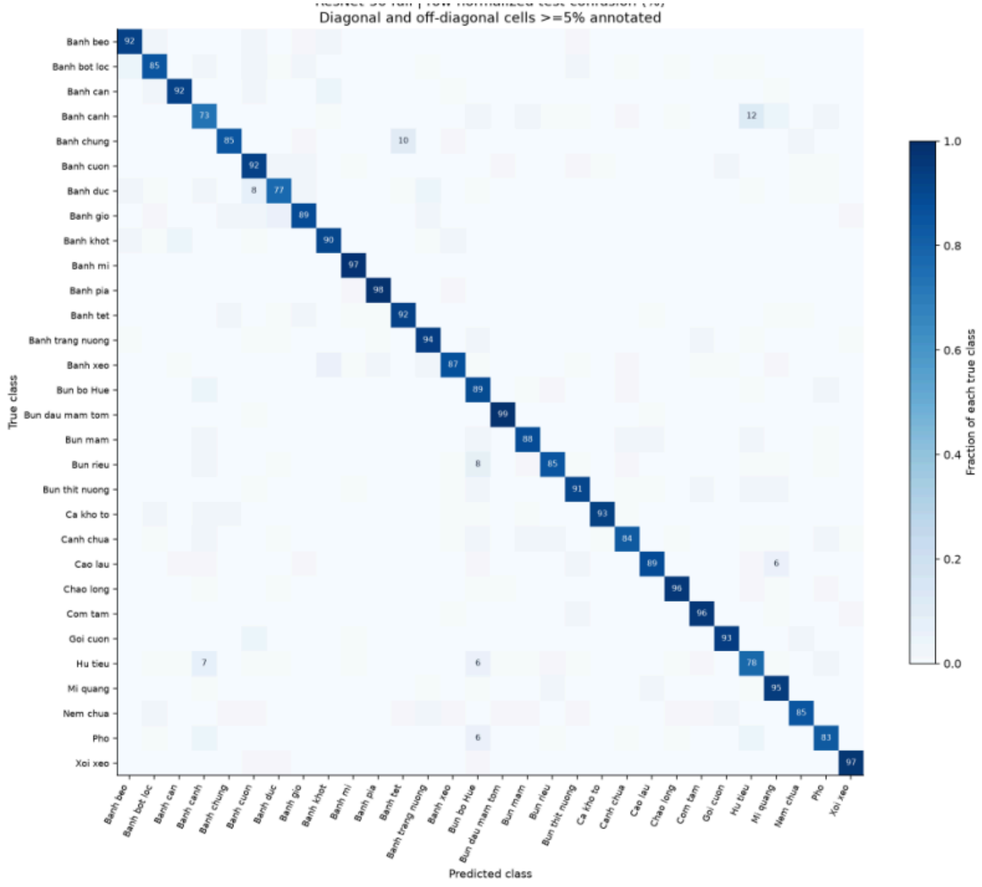
Per-class F1: where fine-tuning helps



Full fine-tuning improves F1 scores across all 30 food categories, with the largest gains in Bun rieu, Banh duc, and Hu tieu, although Banh canh remains the most challenging class despite significant improvement.

04 / ERROR ANALYSIS

Confusion matrix & persistent errors



igure 9: Row-normalized confusion matrix for the validation-selected full fine-tuning check oint. Entries annotated in the heatmap are percentages.

Banh canh → Hu tieu

22 of 191 images (11.5%) in full seed 44.

Bun rieu → Bun bo Hue

Reduced from 52/227 (frozen) to 19/227 (full).

Confusions reflect overlapping ingredients and presentation.

04 / ERROR ANALYSIS

Qualitative evidence & Grad-CAM



Correct (#2250)



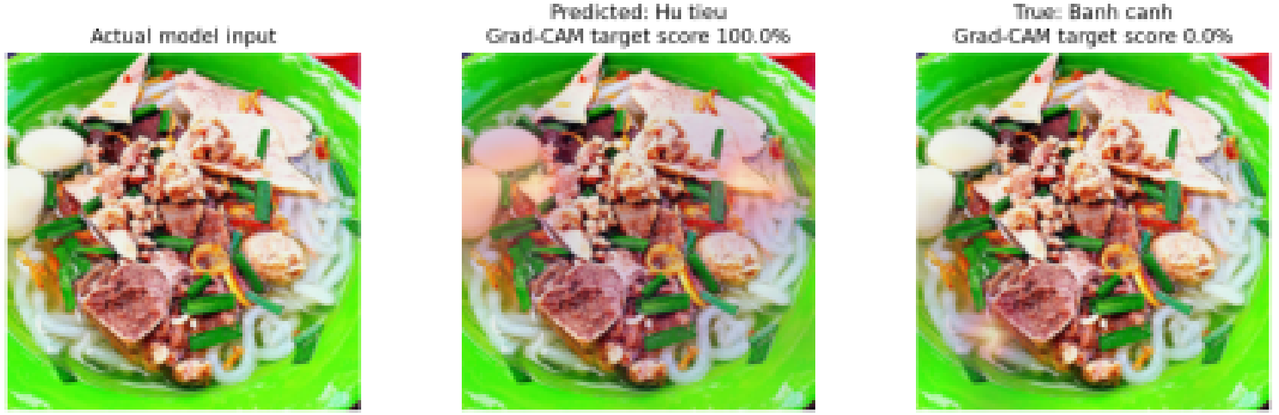
Difficult (#4165)



Failure (#568)

Correct / difficult / failed predictions

> **Interpretation**
>
> High-confidence errors remain. Grad-CAM indicates salient regions; it does not prove the cause of misclassification.



Grad-CAM on Banh canh → Hu tieu

04 / DISCUSSION

Limitations & future work

Dataset leakage

pHash thresholds can miss near-duplicates or group distinct images.

Generalization

Only one cleaned split; unseen sources and real user photos are untested.

Experimental coverage

Three seeds for ResNet, one seed for CNN; BN behavior changes with full adaptation.

Next experiments

Compare center crops vs aspect-preserving padding; audit pHash false positives.

Additional work: investigate Banh canh–Hu tieu errors and validate on new image sources.

04 / CONCLUSION

Conclusion & Q&A

89.08%	+0.1202	2.06×
Full ResNet-50 accuracy	Macro-F1 vs frozen	Training-time cost

- Fine-tuning ImageNet features substantially improves 30VNFoods recognition.
- Benefits are consistent across three paired seeds, but cost more training compute.
- Visually similar dishes and possible preprocessing artifacts remain important challenges.

Thank you! Questions?